

KAMPALA INTERNATIONAL UNIVERSITY, WESTERN CAMPUS.  
FACULTY OF BIOMEDICAL SCIENCES  
DEPARTMENT OF HUMAN ANATOMY  
MBChB 2.1 CAT OF SEMESTER EXAMINATION (CAT)

DATE: 1<sup>st</sup> OCTOBER 2021

TIME: 120 minutes

INSTRUCTIONS: Attempt all questions. ONLY REG. NO. IS PERMITTED TO BE WRITTEN ON QUESTION PAPER & TO BE SUBMITTED.

REGISTRATION NO: \_\_\_\_\_

SECTION A: MULTIPLE CHOICE QUESTIONS

ANSWERS ARE NOT ALLOWED IN THE PAPER EXAM....

PLEASE ANSWER ALL THE QUESTIONS IN THE ANSWER BOOKLET  
SECTION A: There are 60 questions in this section, select the correct and appropriate option from letters A to D or E.

1. Which ligaments associate with the lamina of vertebral arch adjacently?
  - a) Posterior longitudinal ligament.
  - b) Supraspinal ligament.
  - c) Anterior longitudinal ligament.
  - d) Interspinal ligament.
  - e) None of above.
2. The first sternocostal joint belongs to the variety of:
  - a) Syndesmosis.
  - b) Symphysis.
  - c) Synchondrosis.
  - d) Synovial.
  - e) None of above.
3. The neck of the first rib is related to:
  - a) Scalenus anterior.
  - b) Stellate ganglion.
  - c) Subclavius.
  - d) Scalenus medium.
  - e) Scalenus posterior.
4. The manubrium gives attachment to the following muscles, except:
  - a) Subclavius.
  - b) Sternohyoid.
  - c) Sternocleidomastoid.
  - d) Pectoralis major.
  - e) Sternothyroid.
5. Concerning thoracic vertebrae, find the false statement:
  - a) Have facets on their bodies for the heads of ribs.
  - b) Have articular processes shaped to allow rotation of the vertebral column.
  - c) Articulate with the rib numbered next below.
  - d) Do not articulate with the rib of the same number.
  - e) Have facets on their transverse processes for tubercles of ribs.
6. The normal quiet expiration is achieved by contraction of which of the following structures?
  - a) Elastic tissue in the lungs and thoracic wall.
  - b) Serratus posterior superior muscles.
  - c) Pectoralis minor muscles.
  - d) Serratus anterior muscles.
  - e) Diaphragm.
7. In the thorax:
  - a) The inlet is bounded by the scapula, clavicles and sternum.
  - b) The inferior border of the superior mediastinum is level with T4.
  - c) The right phrenic nerve passes through the oesophageal opening.
  - d) The left phrenic nerve passes through the oesophageal opening.
  - e) The thoracic duct passes through the caval opening.
8. The following functions concerning the thoracic cage are true, except:
  - a) Protects vital organs within the thorax.
  - b) Supports thorax during respiration.
  - c) Supports shoulder girdles and upper limbs.
  - d) Provides attachments points for back, chest and shoulder muscles.
  - e) None of the above.

9. Which of the following statements best describes the diaphragm?
  - a) It is attached to the xiphoid.
  - b) It is attached to the upper three lumbar vertebrae on the left.
  - c) The oesophagus passes through it at the level of T8.
  - d) The left phrenic nerve pierces the diaphragm with the inferior vena cava.
  - e) Congenital Bochdalek hernia occurs anteriorly.
10. The esophageal opening in the diaphragm transmits all except:
  - a) Lymphatics.
  - b) Right phrenic nerve.
  - c) Vagal trunk.
  - d) Esophageal branches of gastric artery.
  - e) None of above.
11. Which of the following do not penetrate the diaphragm?
  - a) Aorta.
  - b) Inferior vena cava.
  - c) Left phrenic nerve.
  - d) Right phrenic nerve.
  - e) Esophagus.
12. Opposite which thoracic vertebra the thoracic duct inclines to the left?
  - a) 1<sup>st</sup>.
  - b) 3<sup>rd</sup>.
  - c) 5<sup>th</sup>.
  - d) 7<sup>th</sup>.
  - e) 9<sup>th</sup>.
13. When you inhale, the diaphragm:
  - a) Relaxes.
  - b) Does not move.
  - c) Moves upward.
  - d) Moves downward.
  - e) None of above.
14. The thoracic duct ends by opening into:
  - a) Right subclavian vein.
  - b) Left jugular vein.
  - c) Left venous angle.
  - d) Left subclavian vein.
  - e) Right venous angle.
15. The inferior margins of the pleurae are in correspondence with which rib at midaxillary line?
  - a) 7<sup>th</sup> rib.
  - b) 8<sup>th</sup> rib.
  - c) 9<sup>th</sup> rib.
  - d) 10<sup>th</sup> rib.
  - e) 11<sup>th</sup> rib.
16. The following statements belong to the parietal pleura, except?
  - a) Costal pleura.
  - b) Diaphragmatic pleura.
  - c) Mediastinal pleura.
  - d) Cupula of pleura.
  - e) None of above.
17. Mediastinal surface of right lung is related to the following structures, except:
  - a) Azygos vein.
  - b) Superior vena cava.
  - c) Thoracic duct.
  - d) Trachea.
  - e) None of above.
18. The Bronchopulmonary lymph nodes are in:
  - a) In the hilum of each lung.
  - b) In the angle between the trachea and principle bronchi.
  - c) In the lobes of each lung.
  - d) In the pulmonary segment
  - e) None of above.
19. The phrenic nerve descends:
  - a) In front of the root of lung.
  - b) Behind the root of lung.
  - c) In front of the subclavian vein.
  - d) Behind the subclavian artery.
  - e) Behind the Scalenus anterior.
20. The thoracic duct begins at:
  - a) Right lumbar trunk.
  - b) Left lumbar trunk.
  - c) Intestinal trunk.
  - d) Left bronchomediastinal trunk.
  - e) None of above.
21. The left coronary artery supplies blood to:
  - a) Right atrium.
  - b) Anterior wall of the left ventricle.
  - c) Posterior wall of the right ventricle.
  - d) All of the above.
  - e) None of the above.
22. The tributaries of the coronary sinus are:
  - a) Great cardiac vein.
  - b) Middle cardiac vein.
  - c) Small cardiac vein.
  - d) All of the above.
  - e) None of the above.

23. A 5-year-old girl is brought to the emergency department because of difficulty breathing (dysnea), palpitations, and shortness of breath. Doppler study of the heart reveals an atrial septal defect (ASD). This malformation usually results from incomplete closure of which of the following embryonic structures?
- Ductus arteriosus.
  - Ductus venosus.
  - Sinus venarum.
  - Truncus arteriosus.
  - None of the above.
24. A newborn baby is readmitted to the hospital with hypoxia and upon testing is found to have pulmonary stenosis, transposition of aorta, interventricular septal defect, and hypertrophy of the right ventricle. Which of the following is best described by these symptoms?
- ASD.
  - Patent ductus arteriosus.
  - Tetralogy of Fallot.
  - Aortic stenosis.
  - Coarctation of aorta.
25. Consider these statements about the heart, which one is incorrect?
- The right lateral border is formed entirely by the right atrium.
  - The left atrium is visible when the anterior surface of the heart is viewed.
  - The inferior surface is formed by the left and right ventricles, together with the right atrium.
  - The base of the heart is directly related to the surface of the diaphragm.
  - The pulmonary trunk arises anterior to the aorta.
26. Which statement about the conducting system of the heart is correct?
- The sinoatrial node lies in the crista terminalis, next to the entrance of the inferior vena cava.
  - The atrioventricular node lies next to the opening of the coronary sinus.
  - The bundle of His divides in the membranous part of the interventricular septum.
  - The right bundle branch has anterior and posterior divisions.
  - The sinoatrial node is supplied by the right coronary artery in 40% of cases.
27. Which statement about the valves of the heart is correct?
- Both the pulmonary and aortic valves are bicuspid.
  - The 1st heart sound corresponds to closure of the aortic and pulmonary valves.
  - Closure of the aortic valve is best heard in the left 2nd intercostal space.
  - The chorda tendinae tether the atrioventricular valves to pectinate muscle directly.
  - The chorda tendinae prevent eversion of the atrioventricular valves.
28. Which statement about the primitive aortic arches is correct?
- The first arches persist.
  - The second arches become the carotids.
  - The third arches disappear.
  - The fourth arch links up with the descending aorta on the left side.
  - The fifth arches become the pulmonary arteries.
29. Separation of the oesophagus and trachea is completed by:
- Fusion of the palatal shelves.
  - Establishment of the branchial arches.
  - Fusion of the oesophagotracheal ridges to form a septum.
  - Cell necrosis in the lateral margins of the foregut.
  - Expansion of the respiratory diverticulum in a caudal direction.
30. Which heart valve has three cusps?
- Aortic.
  - Mitral.
  - Pulmonary.
  - Pulmonary and aortic.
  - None of the above.



31. The respiratory diverticulum that develops into the lungs appears in \_\_\_\_\_ old embryo.
- 1 week.
  - 2 weeks.
  - 3 weeks.
  - 4 weeks.
  - 5 weeks.
33. Which of the following forms ligamentum teres hepatis after birth?
- Hepatic portal vein.
  - Umbilical artery.
  - Umbilical vein.
  - Hepatic vein.
  - Hepatic artery.
35. Arrange the following from earliest to the latest stage in lung maturation.
- Canalicular- Terminal sac- Alveolar- Pseudo glandular.
  - Canalicular- Terminal sac- Pseudo glandular- alveolar.
  - Pseudo glandular- Terminal sac- Canalicular- Alveolar.
  - Canalicular- Pseudo glandular- Terminal sac- Alveolar.
  - Pseudo glandular- Canalicular- Terminal sac- Alveolar.
37. The following veins drain into the left brachiocephalic vein, except:
- Inferior thyroid vein.
  - Azygos vein.
  - Left vertebral vein.
  - Left internal thoracic vein.
  - Thymic vein.
39. The following statements about the projection of the inferior margins of lungs are true, except:
- Midclavicular line at 6<sup>th</sup> rib.
  - Midaxillary line at 8<sup>th</sup> rib.
  - Scapular line at 10<sup>th</sup> rib.
  - Midaxillary line at 10<sup>th</sup> rib.
  - None of the above.
41. The branches of the ascending aorta include:
- Brachiocephalic trunk.
  - Left common carotid artery.
  - Left subclavian artery.
  - Thyroid ima artery.
  - Coronary artery.
43. An infant is born with a large defect in the central tendon of the diaphragm. This condition is most likely the result of malformation of which of the following structures?
- Dorsal mesentery of the esophagus.
  - Ventral mesentery of the gut tube.
  - Pleuroperitoneal folds.
  - Septum transversum.
  - Cervical somites.
32. Which of the following gives rise to the left ventricle?
- Primitive ventricle.
  - Proximal part of bulbous cordis.
  - Conus cordis.
  - Truncus arteriosus.
  - Sinus venosus.
34. Hyaline membrane disease of the newly born is due to:
- Insufficient surfactant.
  - Too much surfactant.
  - Thickened alveolar wall.
  - Very thin alveolar wall.
  - Reduced number of alveoli.
36. Tetralogy of Fallot includes:
- Persistent atrioventricular canal patent ostium primum.
  - Stenosis of the aortic valve and atrial septal defect.
  - Stenosis of the aortic valve and interventricular septal defect.
  - Pulmonary infundibular stenosis and interventricular septal defect.
  - Interventricular septal defect only.
38. Landmarks of the trachea are:
- Thyroid cartilage to sternal notch.
  - Hyoid bone to sternal angle.
  - Cricoid cartilage to sternal angle.
  - Thyroid cartilage to sternal angle.
  - Cricoid cartilage to sternal notch.
40. A specific coronary vessel that lies in the anterior interventricular sulcus is the:
- Posterior interventricular artery.
  - Right coronary artery.
  - Left coronary artery.
  - Right marginal artery.
  - None of the above.
42. Which is not found in the posterior mediastinum?
- Descending thoracic aorta.
  - Thoracic duct.
  - Phrenic nerves.
  - Azygos vein.
  - Lymph nodes
44. Deep inguinal ring is a defect in the:
- Aponeurosis of external oblique muscle of abdomen.
  - Transversalis fascia.
  - Aponeurosis of internal oblique muscle of abdomen.
  - Aponeurosis of transversus abdominis muscle.
  - None of the above.

45. Regarding the rectus sheath:
- It has both anterior and posterior layers running same lengths.
  - It is contributed to by the aponeurosis of the external oblique, internal oblique and the transverse abdominis.
  - In the midline it forms the linea alba which runs with the same consistent strength throughout its entire length.
  - All of the above.
  - None of above.
47. Indirect inguinal herniae, find the FALSE statement:
- Pass through the Hesselbach's triangle.
  - Are more common in males.
  - Are often congenital.
  - Pass through the deep inguinal ring.
  - Lies inferior and lateral to the pubic tubercle.
49. Which of the following statements regarding the divisions of the mediastinum is true?
- The superior mediastinum lies above the level of the T3/T4 intervertebral disc.
  - The inferior mediastinum is conventionally divided into three parts.
  - The descending thoracic aorta is entirely in the middle mediastinum.
  - The pericardial sac and its contents form the anterior mediastinum.
  - The phrenic nerves run in the anterior compartment.
46. The linea alba is:
- The white line around the iris of the elderly people.
  - The light reflex of the tympanic membrane.
  - Formed by the fusion of eighth, ninth and tenth costal cartilage.
  - A reflection of the inguinal ligament.
  - None of above.
48. The following are related to the posterior surface of the manubrium sterni except:
- Left brachiocephalic vein.
  - The subclavius muscle.
  - The thymus remnants.
  - Sternohyoid and sternothyroid muscle.
  - Pectoralis major muscle.
50. Indirect inguinal herniae, find the FALSE statement:
- Pass through the Hesselbach's triangle. *marked*
  - Are more common in males.
  - Are often congenital.
  - Pass through the deep inguinal ring.
  - Lies inferior and lateral to the pubic tubercle.

### SECTION B: STRUCTURED QUESTIONS

ANSWER THE QUESTIONS IN ITS RESPECTIVE SPACE IN THE ANSWER SHEET

Instruction: This section contains 20 questions; each question should take not more than three minutes.

- The laminae and pedicles of each vertebra make up the vertebral arch.
- The intervertebral foramen is formed by notched areas on two articulated vertebrae, and provides passages through which spinal nerves to leave the spinal cord.
- The annulus fibrosus is the outer, fibrocartilage layer of an intervertebral disc.
- The inferior vena cava returns blood from body regions below the diaphragm.
- The inner walls of the ventricles of the heart are marked by ridges of muscle called crossbars of flesh, which
- In many (but not most) of the population, the circumflex artery continues until it reaches the interventricular septum, where it forms a \_\_\_\_\_.
- The \_\_\_\_\_ prevent backflow into the atria when the ventricles contract together.

8. If one of the heart's valves becomes stiff or narrow, slowing blood flow, the condition is called \_\_\_\_\_.
9. Serous membranes which surround the lungs are called the \_\_\_\_\_.
10. Several organs lie outside the peritoneal cavity, between the parietal peritoneum and the dorsal abdominal wall. These are referred to as \_\_\_\_\_.

#### SECTION C: ESSAY QUESTIONS

ANSWER THE QUESTIONS IN ITS RESPECTIVE SPACE IN THE ANSWER SHEET

Instruction: This section contains 2 questions; each question should take not more than ten minutes.

1. Write a short essay on the external and internal features of the right atrium
2. Write short notes on the posterior mediastinum, taking into consideration limits and contents.

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FACULTY OF BIOMEDICAL SCIENCES  
DEPARTMENT OF HUMAN ANATOMY  
BMS 2.1 MID- SEMESTER EXAMINATION (CAT)

DATE: 29<sup>th</sup> MARCH 2021

TIME ALLOWED: 150 minutes

GENERAL INSTRUCTIONS:

- A. ATTEMPT ALL QUESTIONS
- B. ONLY REG. NO. IS PERMITTED TO BE WRITTEN ON QUESTION PAPER & QUESTION PAPER BE SUBMITTED.
- C. ANSWERS ARE NOT ALLOWED IN THE EXAM QUESTION PAPER.
- D. ANSWER ALL QUESTIONS IN THE ANSWER SHEETS & BOOKLETS PROVIDED

REGISTRATION NO: \_\_\_\_\_

SECTION A: MULTIPLE CHOICE QUESTIONS

SECTION A INSTRUCTIONS:

SECTION A: There are 100 compulsory questions in this section, select the correct/appropriate option from letters A to D or E. Each question should take one minute and is worth one mark.

1. Concerning the thoracic cage, which of the following is false
  - a) It has a superior thoracic aperture which is bounded anteriorly by the manubrium
  - b) The trachea, enters it via the inferior thoracic aperture
  - c) It is bounded laterally by ribs and costal cartilages
  - d) Superior thoracic aperture is oblique and kidney shaped
  - e) It is larger/wider in males than in females
2. True ribs are
  - a) Ribs that connect directly to the sternum via their costal cartilages
  - b) Ribs that connect indirectly to the sternum via their costal cartilages
  - c) They are ribs 1 to 7 ✓
  - d) A & C only
  - e) B & C only
3. Cardiopulmonary resuscitation involves chest compressions to manually start the heart, this technique is possible because
  - a) The thoracic cage is dome shaped
  - b) Costal cartilages provide elasticity & flexibility for chest compression
  - c) The heart is an elastic organ
  - d) All of the above
  - e) B & C only



except

- a) 1<sup>st</sup> rib
  - b) 2<sup>nd</sup> rib
  - c) 10<sup>th</sup> rib
  - d) 9<sup>th</sup> rib
  - e) 12<sup>th</sup> rib
5. An exaggerated curvature of the thoracic vertebrae is called
- a) Scoliosis
  - b) Kyphosis
  - c) Lordosis
  - d) Stenosis
  - e) None of the above
6. The midline structure which marks the superior limit of the liver, the central tendon of the diaphragm & also the inferior border of the heart is the
- a) Manubrium
  - b) Body of sternum
  - c) Xyphoid process
  - d) Inferior thoracic aperture
  - e) 1<sup>st</sup> part of duodenum
7. Arterial supply to the thoracic wall comes from the following except
- a) Thoracic aorta ✓
  - b) Abdominal aorta
  - c) Subclavian artery ✓
  - d) Axillary artery ✓
  - e) A & C only
8. Concerning the pleural membranes of the lungs, which of these is incorrect
- a) It has a pleural cavity filled with serous pleural fluid ✓
  - b) The parietal layer is insensitive to pain
  - c) The visceral layer is insensitive to pain
  - d) The visceral layer lines the surface of the lungs
- superiorly into the root of the neck as the cervical pleura
9. The following are the relationships between the vagus & phrenic nerves at the root of the lungs
- a) Vagus crosses anteriorly while phrenic crosses posteriorly
  - b) Phrenic crosses anteriorly while vagus crosses posteriorly
  - c) Vagus crosses superiorly while phrenic crosses inferiorly
  - d) Phrenic crosses superiorly while vagus crosses inferiorly
  - e) None of the above
10. An aspirated particle will likely end in the right lungs because
- a) The right lungs is wider
  - b) The right lungs has 3 lobes
  - c) The right main bronchus is shorter & wider & runs more vertically
  - d) The right lungs is closer to the heart
  - e) The right lungs is pyramid shaped
11. The fibrous pericardium and the parietal layer of the serous pericardium are supplied by the .....
- a) phrenic nerves
  - b) sympathetic trunk
  - c) vagus nerve
  - d) intercostal nerve
  - e) long thoracic nerve
12. The origin of the diaphragm can be divided into ..... parts.
- a) two



- c) three
- d) four
- d) five
- e) six

13. The sternocostal surface of the heart is formed mainly by the .....

- a) right and left atria
- b) right and left ventricles
- c) right atrium and right ventricle
- d) left atrium and left ventricle
- e) none of the above

14. The common cause of defective conduction through the bundle or its branches is called.....

- a. cardiac tamponade ✓
- b. atherosclerosis of the coronary arteries ✓
- c. myocardial infarction ✓
- d. fibrosis of the myocardium
- e. heart burn

15. concerning the apex of the heart, which of the following statements is not true

- a) it is formed by the inferolateral part of the left ventricle
- b) it lies posterior to the left fifth intercostal space in adults
- c) it moves during the cardiac cycle
- d) it is where sounds of mitral valves closure are maximal
- e) it remains motionless throughout the cardiac cycle ✓

16. The following structures are found indenting the left lung except

- a) Aorta
- b) Inferior vena cava
- c) Carotid artery
- d) Cardiac notch

not true about the features of the lungs?

a) An apex, the blunt superior end of the lung ascending above the level of the 1st rib into the root of the neck; the apex is covered by cervical pleura ✓

b) A base, the concave inferior surface of the lung, opposite the apex, resting on and accommodating the ipsilateral dome of the diaphragm. ✓

c) Three lobes, created by a fissure. 3 lobes - Right, 2 lobes - Left lung ✓

d) Three surfaces (costal, mediastinal, and diaphragmatic).

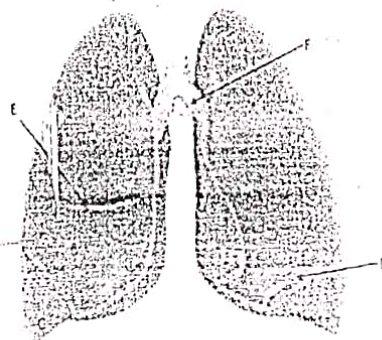
18. Structures indenting the right lungs include

- a) Thoracic duct
- b) Descending aorta
- c) Left common carotid artery
- d) Esophagus ✓

19. The apex of the lung is covered by

- a) Cervical pleura
- b) Diaphragmatic pleura
- c) Costal pleura
- d) Mediastinal pleura

Study the lung below and answer the following questions



20. Name part marked B

- b) Right superior lobe
- c) Right middle lobe
- d) left inferior lobe

21. Name part marked D

- a) Right horizontal fissure
- b) Left horizontal fissure
- c) Left oblique fissure
- d) Right oblique fissure

22. Which ligaments associate with the lamina of vertebral arch adjacently?

- a) Posterior longitudinal ligament.
- b) Supraspinal ligament.
- c) Anterior longitudinal ligament.
- d) Interspinal ligament. *Ligamentum flavum*
- e) None of above

23. The first sternocostal joint belongs to the variety of:

- a) Syndesmosis.
- b) Symphysis.
- c) Synchondrosis.
- d) Synovial.
- e) None of above.

24. The neck of the first rib is related to:

- a) Scalenus anterior.
- b) Stellate ganglion.
- c) Subclavius.
- d) Scalenus medium.
- e) Scalenus posterior.

25. The manubrium gives attachment to the following muscles, except:

- a) Subclavius.
- b) Sternohyoid.
- c) Sternocleidomastoid.
- d) Pectoralis major.
- e) Sternothyroid.

26. Concerning thoracic vertebrae, find the false statement:

- a) Have facets on their bodies for the heads of ribs.

b) Have facets on their transverse processes shaped to allow rotation of the vertebral column.

c) Articulate with the rib numbered next below.

d) Do not articulate with the rib of the same number.

e) Have facets on their transverse processes for tubercles of ribs.

27. The following functions concerning the thoracic cage are true, except:

- a) Protects vital organs within the thorax.
- b) Supports thorax during respiration.
- c) Supports shoulder girdles and upper limbs.
- d) Provides attachments points for back, chest and shoulder muscles.
- e) None of the above.

28. In the thorax:

- a) The inlet is bounded by the scapula, clavicles and sternum.
- b) The inferior border of the superior mediastinum is level with T4. ✓
- c) The right phrenic nerve passes through the oesophageal opening.
- d) The left phrenic nerve passes through the oesophageal opening.
- e) The thoracic duct passes through the caval opening.

29. Landmarks of the trachea are:

- a) Thyroid cartilage to sternal notch.
- b) Hyoid bone to sternal angle.
- c) Cricoid cartilage to sternal angle.
- d) Thyroid cartilage to sternal angle.

notch.

30. The first seven pairs of ribs attach to the sternum by the costal cartilage and are called:
- a) Floating ribs.
  - ☒ b) True ribs.
  - c) False ribs.
  - d) Peculiar ribs.
  - e) None of the above.
31. The following statements concerning the thoracic vertebrae are false, except:
- a) They have transverse foramen. ✓
  - ☒ b) They have the end of the spinous process bifurcated. ✗
  - c) They have costal fovea on the vertebral body laterally. ✓
  - d) The articular facets are relatively sagittal. ✓
  - e) None of the above.
32. Which statement about the ribs is accurate?
- a) The head of a typical rib articulates with the transverse costal facet of its own vertebra.
  - b) The 1st rib has the subclavian artery as an inferior relation. ✗
  - c) The costotransverse ligaments pass from the transverse process of the vertebra to the head of the rib.
  - d) The intercostal neurovascular bundle has the intercostal vein as the most inferior structure.
  - ☒ e) None of the above.
33. Which of the following statements about the surface anatomy of the chest is correct?
- a) The manubrium overlies the aortic arch at the level of T4-T5.
  - b) The body of the sternum overlies the heart at the level of T5-T8.
  - c) The most prominent vertebral spinous process encountered when palpating the spine is that of C6.
  - d) The manubriosternal joint is at the level of the junction of T3 and T4. ✗
  - e) The xiphisternal joint is at the level of T8. ✗
34. Which of the following structures regulate the flow of material into the colon?
- a) Pyloric sphincter.
  - b) Cardiac sphincter.
  - ☒ c) Ileo-caecal sphincter.
  - d) Treitz ligament.
  - e) Appendix.
35. Which organ becomes retroperitoneal during rotation of the gut tube?
- a) Duodenum.
  - ☒ b) Kidney.
  - ☒ c) Spleen.
  - d) Stomach.
  - e) Transverse colon.
36. True statements concerning the ileum include:
- a) It usually has more mesenteric fat than the jejunum.
  - b) Its mesenteric blood vessels possess more arcades than in the mesenteric vessels of the jejunum.
  - c) It is usually located inferior to the jejunum.



- e) Both (only) B and C
37. Which one of the following structures is usually secondarily retroperitoneal?
- a) Adrenal (suprarenal) gland
  - b) Sigmoid colon
  - c) Spleen
  - d) Ileum
  - e) Second part of the duodenum
38. Which one of the following structures is most likely to be located posterior to the second part of the duodenum?
- a) Abdominal aorta
  - b) Neck of gallbladder
  - c) Portal vein
  - d) Hilus of right kidney
  - e) Transverse colon
39. The gastroduodenal artery:
- a) Is usually a branch of the superior mesenteric artery
  - b) Courses anterior to the first part of the duodenum
  - c) Terminates by dividing into superior pancreaticoduodenal and right gastroepiploic arteries
  - d) All the above
  - e) Both (only) b and c
40. The folds of the mucosa of the jejunum are called
- a) Haustra
  - b) Plicae longitudinales
  - c) Rugae
  - d) Plicae circulares
  - e) Gastric pits
41. Most of the duodenum is "primarily" retroperitoneal:
- a) True
  - b) Only in females
  - c) Only in males
  - d) Only in monkeys
  - e) False
42. Which ligaments associate with the lamina of vertebral arch adjacently?
- a) Posterior longitudinal ligament.
  - b) Supraspinal ligament.
  - c) Anterior longitudinal ligament.
  - d) Interspinal ligament.
  - e) None of above
43. The first sternocostal joint belongs to the variety of:
- a) Syndesmosis.
  - b) Symphysis.
  - c) Synchondrosis.
  - d) Synovial.
  - e) None of above.
44. The neck of the first rib is related to:
- a) Scalenus anterior.
  - b) Stellate ganglion.
  - c) Subclavius.
  - d) Scalenus medium.
  - e) Scalenus posterior.
45. The manubrium gives attachment to the following muscles, except:
- a) Subclavius.
  - b) Sternohyoid.
  - c) Sternocleidomastoid.
  - d) Pectoralis major.
  - e) Sternothyroid.
46. Concerning thoracic vertebrae, find the false statement:
- a) Have facets on their bodies for the heads of ribs.
  - b) Have articular processes shaped to allow rotation of the vertebral column.
  - c) Articulate with the rib numbered next below.
  - d) Do not articulate with the rib of the same number.
  - e) Have facets on their transverse processes for tubercles of ribs.



concerning the thoracic cage are true, except:

- a) Protects vital organs within the thorax.
- b) Supports thorax during respiration.
- c) Supports shoulder girdles and upper limbs.
- d) Provides attachments points for back, chest and shoulder muscles.
- e) None of the above.

48. In the thorax:

- a) The inlet is bounded by the scapula, clavicles and sternum.
- b) The inferior border of the superior mediastinum is level with T4.
- c) The right phrenic nerve passes through the oesophageal opening.
- d) The left phrenic nerve passes through the oesophageal opening.
- e) The thoracic duct passes through the caval opening.

49. Landmarks of the trachea are:

- a) Thyroid cartilage to sternal notch.
- b) Hyoid bone to sternal angle.
- c) Cricoid cartilage to sternal angle.
- d) Thyroid cartilage to sternal angle.
- e) Cricoid cartilage to sternal notch.

50. The first seven pairs of ribs attach to the sternum by the costal cartilage and are called:

- a) Floating ribs.
- b) True ribs.
- c) False ribs.
- d) Peculiar ribs.
- e) None of the above.

concerning the thoracic vertebrae are false, except:

- a) They have transverse foramen.
- b) They have the end of the spinous process bifurcated.
- c) They have costal fovea on the vertebral body laterally.
- d) The articular facets are relatively sagittal.
- e) None of the above.

52. Which statement about the ribs is accurate?

- a) The head of a typical rib articulates with the transverse costal facet of its own vertebra.
- b) The 1st rib has the subclavian artery as an inferior relation.
- c) The costotransverse ligaments pass from the transverse process of the vertebra to the head of the rib.
- d) The intercostal neurovascular bundle has the intercostal vein as the most inferior structure.
- e) None of the above.

53. Which of the following statements about the surface anatomy of the chest is correct?

- a) The manubrium overlies the aortic arch at the level of T4-T5.
- b) The body of the sternum overlies the heart at the level of T5-T8.
- c) The most prominent vertebral spinous process encountered when palpating the spine is that of C6.

- at the level of the junction of T3 and T4.
- e) The xiphisternal joint is at the level of T8.
54. Which of the following structures regulate the flow of material into the colon?
- Pyloric sphincter.
  - Cardiac sphincter.
  - Ileo-caecal sphincter.
  - Treitz ligament.
  - Appendix.
55. Which organ becomes retroperitoneal during rotation of the gut tube?
- Duodenum.
  - Kidney.
  - Spleen.
  - Stomach.
  - Transverse colon.
56. True statements concerning the ileum include:
- It usually has more mesenteric fat than the jejunum.
  - Its mesenteric blood vessels possess more arcades than in the mesenteric vessels of the jejunum.
  - It is usually located inferior to the jejunum.
  - All the above
  - Both (only) B and C
57. Which one of the following structures is usually secondarily retroperitoneal?
- adrenal (suprarenal) gland
  - sigmoid colon
  - spleen
- e) second part of the duodenum.
58. Which one of the following structures is most likely to be located posterior to the second part of the duodenum?
- abdominal aorta
  - neck of gallbladder
  - portal vein
  - hilus of right kidney
  - transverse colon
59. The gastroduodenal artery:
- is usually a branch of the superior mesenteric artery
  - courses anterior to the first part of the duodenum
  - terminates by dividing into superior pancreaticoduodenal and right gastroepiploic arteries
  - all the above
  - both (only) B and C
60. The folds of the mucosa of the jejunum are called
- haustra
  - plicae longitudinales
  - rugae
  - plicae circulares
  - gastric pits
61. Indirect Inguinal hernias are more common in males than in females because
- Males work harder than females
  - Descent of testes creates a canal through which structures can pass
  - Males have a longer pelvic canal compared to females
  - Males eat more than females
  - None of the above

62. A patient presents the following symptoms, pain particularly when coughing or stooping, change in bowel habit, constipation, burning sensation in groin, scrotal swelling. As a clinician, your best possible guess is

- a. Direct femoral hernia
- b. Direct inguinal hernia
- c. Indirect femoral hernia
- d. Indirect inguinal hernia
- e. Saphenous varix

63. A patient presents with a slight bulging in the groin and on palpation of the deep inguinal ring you observed that the hernia still protrudes even as you are pressing it to retract into the abdomen, your best possible guess is

- a. Direct inguinal hernia
- b. Indirect inguinal hernia
- c. Direct femoral hernia
- d. Indirect femoral hernia
- e. Groin pull

64. Which of the following is NOT true concerning skeletal development?

- A. Membrane bones include the clavicle and the bones of the cranial vault and face.
- B. During membrane bone formation, no cartilaginous model is formed.
- C. Endochondral bone formation involves calcification of cartilage, which is later replaced by true bone.
- D. Membrane bone lacks periosteum.

characterized by:

- A. ectrodactyly
- B. brachydactyly
- C. syndactyly
- D. polydactyly
- E. more than one of the above

66. The sphenoid bone is derived from which portion of the embryonic skull?

- A. cartilaginous neurocranium
- B. membranous neurocranium
- C. cartilaginous viscerocranium
- D. membranous viscerocranium
- E. branchial cartilage

67. Which of the following is NOT true with respect to the development and growth of bones:

- A. Primary centers of ossification appear before the end of the third month of development.
- B. Almost all secondary ossification centers appear after birth.
- C. Longitudinal growth of a long bone occurs in the areas of the epiphyseal plates.
- D. Membrane bones can be recognized by their typical histological structure.

68. Achondroplasia is characterized by all of the following EXCEPT:

- A. inferior sexual development
- B. short limbs
- C. average intelligence
- D. broad, flat face
- E. undershot jaw



59. Digits that are fused together are called.

- A. syndactyly
- B. polydactyly
- C. dichuris
- D. amelus
- E. none of the above

70. What gene or factor in the dorsal ectoderm is responsible for the dorsoventral axis formation of the limbs?

- A. Wnt-7a
- B. En-1
- C. Hox
- D. FGF-8

71. A newborn baby is found to have her 3rd and 4th digits fused. What is the term for this condition?

- A. Ectrodactyly
- B. Brachydactyly
- C. Polydactyly
- D. Syndactyly
- E. Hemimelia

72. The following give rise to the skeletal system except

- A. Paraxial mesoderm
- B. Intermediate mesoderm
- C. Somatic layer of lateral plate mesoderm
- D. Neural crest

73. The mesoderm that forms segmented blocks of tissues on each side of the neural tube

- A. Paraxial
- B. Intermediate
- C. Lateral plate
- D. Somites

74. Which region of a somitomere develops into ribs and vertebrae?

A. Somite

B. Sclerotome

C. Myotome

D. Somatome

75. Development of the skeletal system begins at around..... week of development

- A. 30<sup>TH</sup>
- B. 12<sup>TH</sup>
- C. 20<sup>TH</sup>
- D. 3<sup>RD</sup>

76. Which structure mainly contributes to the formation of the of bones of the pelvic and pectoral girdles and the long bones of the limbs

- A. Intermediate mesoderm
- B. Paraxial mesoderm
- C. Somatic mesoderm
- D. Ectoderm

77. Which structure mainly participates in the formation of the bones of the face and skull

- A. Ectoderm
- B. Somatic mesoderm
- C. Intermediate mesoderm
- D. Neural crest
- E. Myotome

78. In some bones, mesenchyme in the dermis differentiates directly into bone, this process is known as

- A. Dermoid ossification
- B. Endochondral ossification
- C. Intramembranous ossification
- D. Cartilaginous ossification

79. The first site where bone begins to form in the shaft of a long bone or body of an irregular bone

- A. punctum ossificationis primarium
- B. punctum ossificationis secundarium
- C. epiphyseal ossification center
- D. punctum ossificationis tertium

following events except

- A. Pulmonary stenosis
  - B. Overriding aorta
  - C. Interventricular septal defect
  - D. Hypertrophy of the left ventricle
81. The most prominent fontanelle in the new-born skull
- A. Occipital
  - B. Lambdoidal
  - C. Squamous
  - D. Anterior
  - E. Posterior
82. A defect in the skull where the cranial vault fails to form leading to exposure of the brain tissue
- A. Cranioschisis
  - B. Microcephaly
  - C. Craniosynostosis
  - D. Dwarfism
83. A defect in the skull where the cranial sutures fuse prematurely leading to the malformation of the head
- A. Cranioschisis
  - B. Microcephaly
  - C. Craniosynostosis
  - D. Dwarfism
84. Spina bifida where there is a small gap in the spine, but no opening or sac on the back.
- A. Spina bifida aperta
  - B. Spina bifida cystica
  - C. Spina bifida occulta
  - D. Spina bifida operna
85. Regarding the embryology of vertebrae, which is the true statement
- A. includes a membranous portion & a cartilaginous part
  - B. formed by condensation of caudal half of 1 sclerotome & fusion with cranial half of subjacent sclerotome

development, the ones for the vertebrae become visible from the ventrolateral body wall determined by Hox genes.

- D. During formation of vertebrae mesenchyme in the dermis differentiates directly into bone.
86. An infant is born with a large defect in the central tendon of the diaphragm. This condition is most likely the result of malformation of which of the following structures?
- a) Dorsal mesentery of the esophagus.
  - b) Ventral mesentery of the gut tube.
  - c) Pleuroperitoneal folds.
  - d) Septum transversum.
  - e) Cervical somites
87. A 7 days old baby is diagnosed as having congenital neonatal emphysema, which is caused by collapsed bronchi because of failure of bronchial cartilage development. Bronchial cartilages are derived from which of the following derivations?
- a) Ectoderm.
  - b) Mesoderm.
  - c) Endoderm.
  - d) Neuroectoderm.
  - e) None of above
88. Which statement about the primitive aortic arches is correct?
- a) The first arches persist.
  - b) The second arches become the carotids.
  - c) The third arches disappear.
  - d) The fourth arch links up with the descending aorta on the left side.
  - e) The fifth arches become the pulmonary arteries.

atrium is incorrect?

- a) The crista terminalis separates the smooth walled part from the rough walled part.
- b) The opening of the inferior vena cava into the right atrium is valveless.
- c) It contains the embryological remnant of the foramen ovale.
- d) It receives the coronary sinus.
- e) The smooth walled part is derived from the sinus venosus of the primitive heart tube.

90. Which statement incorrectly describes the venous drainage of the heart?

- a) Venae cordis minimae drain into all four chambers.
- b) The anterior cardiac veins drain into the right atrium.
- c) The coronary sinus lies in the posterior interventricular groove.
- d) The middle cardiac vein is a tributary of the coronary sinus.
- e) The small cardiac vein is a tributary of the coronary sinus

91. Which statement about the primitive aortic arches is correct?

- a) The first arches persist.
- b) The second arches become the carotids.
- c) The third arches disappear.
- d) The fourth arch links up with the descending aorta on the left side.
- e) The fifth arches become the pulmonary arteries

92. Which statement regarding the fetal circulation is correct?

- a) The septum secundum lies to the left of the septum primum.
- b) The ductus arteriosus closes at birth.
- c) The foramen ovale is a defect in the septum primum.
- d) The ligamentum teres is the embryological remnant of the right umbilical vein.
- e) None of the above

93. The valves responsible for preventing backflow of blood from the lungs into the heart is:

- a) Tricuspid valve.
- b) Bicuspid valve.
- c) Aortic semilunar valve.
- d) Pulmonary semilunar valve.
- e) Eustachian valve

94. The following statements regarding the inguinal canal are false, except:

- a) The ilioinguinal nerve enters the canal through the deep ring.
- b) Its posterior wall is formed by the external oblique aponeurosis.
- c) It has a roof formed by lower edges of internal oblique and transverses muscles.
- d) The conjoint tendon lies anterior to the superficial inguinal ring.
- e) None of the above

95. Deep inguinal ring is a defect in the:

- a) Aponeurosis of external oblique muscle of abdomen.
- b) Aponeurosis of internal oblique muscle of abdomen.



- abdominus muscle.
- d) Transversalis fascia.
  - e) None of the above
96. During early and middle fetal life, the testes are located in the:
- a) Inguinal canal.
  - b) Thoracic cavity.
  - c) Pelvic cavity.
  - d) Scrotal.
  - e) None of the above
97. The mucosa layer of the stomach is lined by
- a) Simple Squamous epithelium
  - b) Stratified squamous epithelium
  - c) Simple columnar
  - d) Pseudostratified columnar
  - e) None of the above
98. The following are cells of the
- a) Goblet cells
  - b) Brush cells
  - c) Small granule cells
  - d) Podocytes
  - e) Basal cells
99. The fluid that prevents the collapsing tendency of the lungs is the
- a) Surfactant
  - b) Pleural fluid
  - c) Pericardial fluid
  - d) Seminal fluid
  - e) Serum
100. The cells that form the blood air barrier is the
- a) Type I pneumocytes
  - b) Type II pneumocytes
  - c) Type III pneumocytes
  - d) Type IV pneumocytes
  - e) Type V pneumocytes

## SECTION B

1. Tina was hit by a stray bullet which took a trajectory that travelled through the left 2<sup>nd</sup> intercostals space and ruptured the major blood vessel providing systemic circulation to the rest of the body. Briefly discuss the vessel stating its origin, course, relations and major branches
2. Write a short essay on the external and internal features of the right atrium
3. A surgeon mistakenly cuts the left gastric artery during an abdominal surgical procedure, as a medical student with the knowledge of the blood supply of the stomach; briefly discuss how this will affect the area supplied by the left gastric artery.